

Universal vertex algebras and free field realisations

Simon Wood

The Australian National University

Joint work with Olivier Blondeau-Fournier, Pierre Mathieu, David Ridout and Akihiro Tsuchiya

Outline

- Review of vertex operator algebras and modules
- Free field realisation
- Symmetric polynomials
- Classifying simple minimal model modules

Motivation

Bottom up approach to constructing conformal field theories:

- Pick a vertex operator algebra
- Classify modules in a suitable category
- Compute characters and their modular properties
- Find modular invariants
- Compute fusion products of modules
- Compute correlation functions

Module classification and symmetric polynomials

vertex operator algebras

ideals and quotients

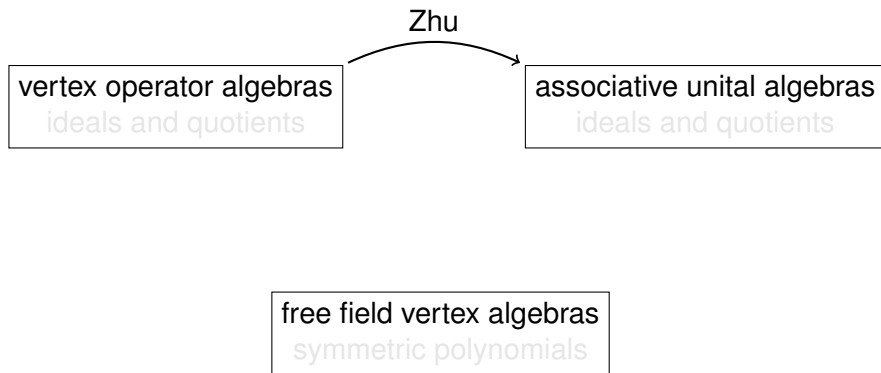
associative unital algebras

ideals and quotients

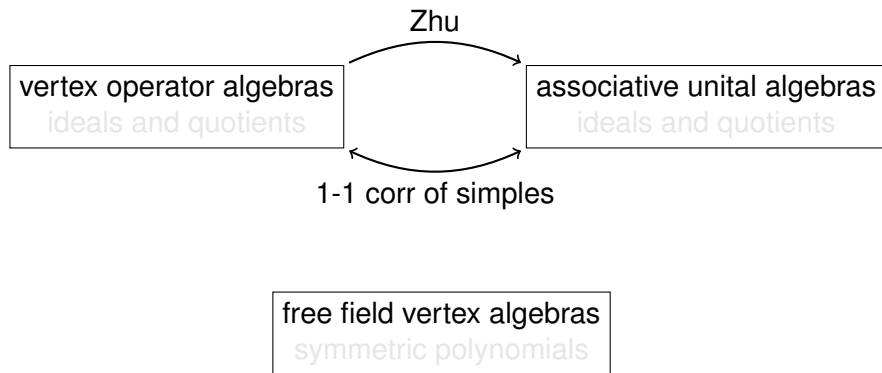
free field vertex algebras

symmetric polynomials

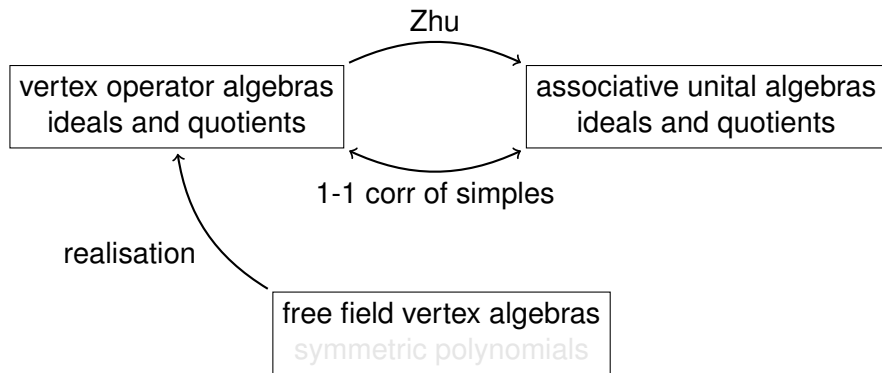
Module classification and symmetric polynomials



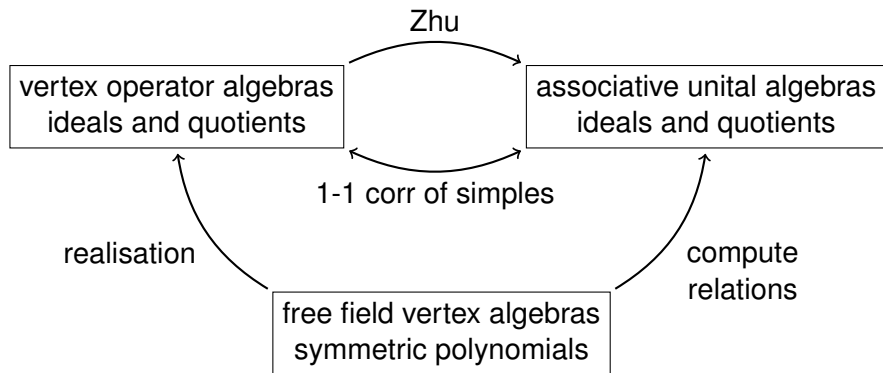
Module classification and symmetric polynomials



Module classification and symmetric polynomials



Module classification and symmetric polynomials



Definition: Vertex operator algebra

A VOA (V, Ω, T, Y) is a collection of data:

- $\mathbb{N}_0$ -graded vector space $V = \bigoplus_{n \geq 0} V[n]$
- vacuum vector $\Omega \in V[0]$
- conformal vector $T \in V[2]$
- linear map (called field map)

$$Y : V \rightarrow \text{End}(V)[[z, z^{-1}]],$$

$$A \mapsto Y(A; z),$$

$$Y(A; z) = \sum_n A_n z^{-n-h_A}, \quad A \in V[h_A]$$

Definition: Vertex operator algebra

The VOA data satisfy the axioms:

- Compatibility with grading: For $A \in V[h_A]$, $A_n : V[m] \rightarrow V[m - n]$
- Vacuum and conformal vectors: $Y(\Omega; z) = 1$,
 $Y(T; z) = T(z) = \sum_n L_n z^{-n-2}$
 $[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}m(m^2 - 1)\delta_{m+n,0}$
- $V[n]$ is eigenspace of L_0 with eigenvalue n .
 L_{-1} is the translation operator
 $[L_{-1}, Y(A; z)] = Y(L_{-1}A; z) = \partial_z Y(A; z)$
- Locality: $\exists N \in \mathbb{N}_0$ such that
 $(z - w)^N [Y(A; z), Y(B; w)] = 0$ in $\text{End}(V)[[z, z^{-1}, w, w^{-1}]]$

Definition: Vertex operator algebra

Prop

For $A, B, C \in V$

$$\left. \begin{aligned} Y(A; z)Y(B; w)C &\in V((z))((w)) \\ Y(B; w)Y(A; z)C &\in V((w))((z)) \\ Y(Y(A; z - w)B; w)C &\in V((w))((z - w)) \end{aligned} \right\} \begin{array}{l} \text{same element of} \\ V[[z, w]][z^{-1}, w^{-1}, (z - w)^{-1}] \\ \text{expanded in 3 domains} \end{array}$$

We identify

$$Y(A; z)Y(B; w) \simeq Y(Y(A; z - w)B; w) \simeq Y(B; w)Y(A; z)$$

and call it the operator product expansion (OPE) of $Y(A; z)$ and $Y(B; w)$.

Example: The universal Virasoro VOA Vir_c

- $\text{Vir}_c = U(\bigoplus_{n \geq 2} \mathbb{C}L_{-n})\Omega$, central charge $c \in \mathbb{C}$,
 $L_n\Omega = 0, n \geq -1$
- vacuum Ω , conformal vector $T = L_{-2}\Omega$
- $Y(L_{-2}\Omega; z) = T(z)$,
- $Y(L_{-n_1-2} \cdots L_{-n_i-2}\Omega; z) = : \left(\frac{\partial^{n_1}}{n_1!} T(z) \right) \cdots \left(\frac{\partial^{n_i}}{n_i!} T(z) \right) :$
- Operator product expansion:

$$T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} + \text{reg}$$

Example: The Heisenberg VOA

Heisenberg Lie algebra: $\mathfrak{h} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C}a_n \oplus \mathbb{C}\mathbf{1}$, $[a_m, a_n] = m\delta_{m+n,0}\mathbf{1}$

Heisenberg VOA:

- $\mathcal{H} = U(\bigoplus_{n \geq 1} \mathbb{C}a_{-n})|0\rangle = \mathbb{C}[a_{-1}, a_{-2}, \dots]|0\rangle$, $a_n|0\rangle = 0, n \geq 0$
- vacuum: $|0\rangle$
- conformal vector: $T = \frac{1}{2}(a_{-1}^2 + \beta a_{-2})|0\rangle$, $c = 1 - 3\beta^2$
- $Y(a_{-1}|0\rangle; z) = a(z) = \sum_n a_n z^{-n-1}$,
 $Y(a_{-n_1-1} \cdots a_{-n_i-1}|0\rangle; z) = : \left(\frac{\partial^{n_1}}{n_1!} a(z) \right) \cdots \left(\frac{\partial^{n_i}}{n_i!} a(z) \right) :$
- Operator product expansion:

$$a(z)a(w) = \frac{1}{(z-w)^2} + \text{reg}$$

Prop

$\text{Vir}_c \rightarrow \mathcal{H}$, $T(z) \mapsto \frac{1}{2} : a(z)^2 : + \frac{\beta}{2} \partial a(z)$,

where $c = 1 - 3\beta^2$, defines an embedding of VOAs. This embedding is called a **free field realisation**.

Definition: VOA modules

- $\mathbb{C}$ -graded vector space $W = \bigoplus_{h \in \mathbb{C}} W[h]$, $W[h]$ generalised L_0 eigenspace, eigenvalue h .
- $Y_W : V \rightarrow \text{End}(W)[[z, z^{-1}]]$ such that for $A, B \in V$ and $C \in W$

$$\left. \begin{aligned} Y_W(A; z)Y_W(B; w)C &\in W((z))((w)) \\ Y_W(B; w)Y_W(A; z)C &\in W((w))((z)) \\ Y_W(Y(A; z-w)B; w)C &\in W((w))((z-w)) \end{aligned} \right\} \begin{array}{l} \text{same element of} \\ W[[z, w]][z^{-1}, w^{-1}, (z-w)^{-1}] \\ \text{expanded in 3 domains} \end{array}$$

VOA module examples

- V is a V -module.
- A proper subspace $I \subset V$ that is a V -module is an ideal of V :
 $Y(V; z)I \subset I((z))$, $Y(I; z)V \subset I((z))$. If $\Omega, T \notin I$, then V/I is a VOA.
- For Vir_c , Virasoro Verma modules $V_h = U(\bigoplus_{n \geq 1} \mathbb{C}L_{-n})\Omega_h$,
 $L_0\Omega_h = h\Omega_h$, $L_n\Omega_h = 0, n \geq 1$.
- For $\mathcal{H}$, Heisenberg Verma modules $F_\lambda = U(\bigoplus_{n \geq 1} \mathbb{C}a_{-n})|\lambda\rangle$,
 $a_0|\lambda\rangle = \lambda|\lambda\rangle$, $a_n|\lambda\rangle = 0, n \geq 1$.

Minimal vs universal Virasoro VOAs

Prop

The universal Virasoro VOA Vir_c has a proper non-trivial ideal if and only if

$$c = 1 - 6 \frac{(p - q)^2}{pq}, \quad p, q \geq 2, \quad \gcd\{p, q\} = 1.$$

This ideal I is unique and is generated by a singular vector of conformal weight $(p - 1)(q - 1)$.

- The simple quotient $\mathbf{M}(p, q) = \text{Vir}_c / I$ is called the Virasoro minimal model VOA.
- A Vir_c -module W is a $\mathbf{M}(p, q)$ -module if and only if I acts trivially: $Y(A; z)W = 0, \forall A \in I$.
- Today's goal: Classify $\mathbf{M}(p, q)$ -modules from Vir_c -module theory.

Example: Yang-Lee model

- central charge $c = -\frac{22}{5}$ Virasoro VOA ($p = 2, q = 5$)
- maximal proper ideal generated by a singular vector χ

$$\chi = (L_{-2}^2 - \frac{3}{5}L_{-4})\Omega$$

$$Y(\chi; z) = : T(z)^2 : - \frac{3}{10} \partial^2 T(z) = \sum_{n \in \mathbb{Z}} \chi_n z^{-n-4}$$

- Evaluate χ_0 on a candidate Virasoro highest weight vector Ω_h .

$$0 = \chi_0 \Omega_h = (L_0^2 + \frac{1}{5}L_0)\Omega_h = h(h + \frac{1}{5})\Omega_h$$

- Conformal weight can only be 0 or $-\frac{1}{5}$.

Zhu's algebra

Previous example was not a fluke. Up to some “technicalities” (codified in Zhu's algebra), VOA module theory is completely characterised by the action of zero modes on highest weight vectors.

Zhu's algebra:

- Zhu's algebra is the associative algebra of zero modes acting on vectors in modules which are annihilated by positive modes.
- Constructible as a vector space quotient of V .
 $\text{Zhu}(V/I) = \text{Zhu}(V)/\text{Zhu}(I)$.
- Simple V -modules $\xleftrightarrow{1:1}$ simple Zhu-modules.

Yang-Lee using Zhu's algebra

- $\text{Zhu}(\text{Vir}_c) = \mathbb{C}[x]$ for all $c \in \mathbb{C}$, where $x = [T]$ is the image of the conformal vector.
- For $c = -\frac{22}{5}$, the image of the singular vector is $[\chi] = f(x) = x(x + \frac{1}{5})$.
- The polynomial $f(x)$ gives an explicit presentation of the minimal model Zhu algebra.
$$\text{Zhu}(\text{M}(2, 5)) = \frac{\mathbb{C}[x]}{\langle f(x) \rangle}$$
- On simple modules over $\text{Zhu}(\text{M}(2, 5))$, x must act by multiplication by a zero of $f(x)$.

Classifying minimal model modules

- Previous example depended heavily on explicit form of the singular vector. Singular vectors are **very** hard to find in general. No general closed formulae are known.
- Goal for the rest of the talk: Find a better way to evaluate zero modes of singular vectors on candidate highest weight vectors.

Singular vectors in the free field realisation

The Heisenberg VOA $\mathcal{H}$ is special, because one can define a field map for vectors in modules, not just vectors in the VOA.

Define

- new operator $\hat{a}$: $[a_n, \hat{a}] = \delta_{n,0}$,
- $e^{\lambda \hat{a}}|\mu\rangle = |\lambda + \mu\rangle$.

Vertex operator (not to be confused with a field in a VOA)

- $Y(|\lambda\rangle; z) = V_\lambda(z) = e^{\lambda \hat{a}} z^{\lambda a_0} \prod_{m \geq 1} \exp\left(\lambda \frac{a_{-m}}{m} z^m\right) \prod_{m \geq 1} \exp\left(-\lambda \frac{a_m}{m} z^{-m}\right)$
- $T(w)V_\lambda(z) = \frac{h_\lambda}{(w-z)^2} V_\lambda(z) + \frac{1}{w-z} \partial V_\lambda(z) + \text{reg}$
- $V_\lambda(z)V_\nu(w) = (z-w)^{\lambda\nu} : V_\lambda(z)V_\nu(w) :$
- $V_\lambda(z) : F_\mu \rightarrow F_{\lambda+\mu}[[z, z^{-1}]]z^{\lambda\mu}$

Singular vectors in the free field realisation

Screening operator:

- If $h_\lambda = 1$ then the field $S(z) = V_\lambda(z)$ is called a screening field.

$$T(w)S(z) = \frac{1}{(w-z)^2}S(z) + \frac{1}{w-z}\partial S(z) + \text{reg} = \partial_z \frac{S(z)}{w-z} + \text{reg}$$

- Zero mode $S_0 = \text{Res}_{z=0} [S(z)]$ is called a screening operator and defines a Virasoro homomorphism:
 $[T(w), S_0] = \text{Res}_{z=w} [T(w)S(z)] = 0$
- **However**, the map $S_0 : F_\mu \rightarrow F_{\mu+\lambda}$ is only well defined if $\lambda\mu \in \mathbb{Z}$.
- If $\lambda\mu \in \mathbb{Z}$ and $S_0|\mu\rangle \neq 0$ then $S_0|\mu\rangle$ is a **singular vector** in $F_{\mu+\lambda}$

The singular vector at minimal model central charges

Let $p, q \geq 2$, $\gcd\{p, q\} = 1$ and

$$\alpha_+ = \sqrt{\frac{2q}{p}}, \quad \alpha_- = -\sqrt{\frac{2p}{q}}, \quad \beta = \alpha_+ + \alpha_-, \quad c = 1 - 3\beta^2 = 1 - 6\frac{(p-q)^2}{pq}.$$

There are two screening fields

$$S_+(z) = V_{\alpha_+}(z), \quad S_-(z) = V_{\alpha_-}(z).$$

The singular vector that generates the ideal of $\mathbf{Vir}_c \subset \mathcal{H}$ by which one must quotient to get the minimal model VOA:

$$\chi = S_+^{[p-1]}|(1-p)\alpha_+\rangle \propto S_-^{[q-1]}|(1-q)\alpha_-\rangle \in \mathbf{Vir}_c \subset \mathcal{H}.$$

The constraints on the minimal model conformal weights are obtained by evaluating the action of the zero mode of χ .

Composing screening operators

To construct the singular vector of $\text{Vir}_c \subset \mathcal{H}$, we must consider higher powers of the screening operator.

$$S_+(z_1) \cdots S_+(z_r) = e^{r\alpha_+ \hat{a}} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{\alpha_+ + a_0} \\ \prod_{m \geq 1} \exp \left(\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m \right) \exp \left(-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m} \right)$$

Prop

If and only if $\mu = \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-$, $k \in \mathbb{Z}$, there exists a unique choice of a cycle Γ such that

$$S_+^{[r]} = \int_{\Gamma} S_+(z_1) \cdots S_+(z_r) dz_1 \cdots dz_r : F_{\frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-} \rightarrow F_{\frac{1+r}{2}\alpha_+ - \frac{k}{2}\alpha_-}$$

is a non-trivial Virasoro homomorphism.

- if $k \geq 0$ then $S_+^{[r]} | \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_- \rangle = 0$.
- if $k < 0$ then $S_+^{[r]} | \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_- \rangle \neq 0$.

Appearance of symmetric polynomials

Action of $S_+^{[r]}$ on $F_{\frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-}$:

$$S_+^{[r]} = e^{r\alpha_+ \hat{a}} \int_{\Gamma} \prod_{i \neq j} \left(1 - \frac{z_i}{z_j}\right)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{k+1} \\ \times \prod_{m \geq 1} \exp\left(\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m\right) \exp\left(-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m}\right) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This integral defines a well known inner product on the ring of symmetric polynomials $\mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$:

$$\langle f, g \rangle_r^t = \int_{\Gamma} \prod_{i \neq j} \left(1 - \frac{z_i}{z_j}\right)^t f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This inner product is symmetric and it vanishes if $\deg(f) \neq \deg(g)$.

Symmetric polynomials

Ring of symmetric polynomials in r variables: $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$

Many well known generators:

- Power sums

$$p_k(z) = \sum_i z_i^k$$

- Elementary symmetric polynomials

$$e_k(z) = \sum_{1 \leq i_1 < \dots < i_k} z_{i_1} \cdots z_{i_k}$$

$$\Lambda_r = \mathbb{C}[p_1, \dots, p_r] = \mathbb{C}[e_1, \dots, e_r]$$

Symmetric polynomials

Many well known bases. Generally indexed by partitions

$$\lambda = [\lambda_1, \lambda_2, \dots].$$

- Power sum basis: $p_\lambda = p_{\lambda_1} p_{\lambda_2} \cdots$
- Elementary symmetric basis: $e_\lambda = e_{\lambda_1} e_{\lambda_2} \cdots$
- Monomial symmetric polynomials:

$$m_\lambda(z) = \sum_{\sigma} z_1^{\lambda_{\sigma(1)}} z_2^{\lambda_{\sigma(2)}} \cdots,$$

where the σ are all distinct permutations of $\lambda = [\lambda_1, \lambda_2, \dots]$.

$$m_{[2,2]}(z_1, z_2) = z_1^2 z_2^2, \quad m_{[2,2]}(z_1, z_2, z_3) = z_1^2 z_2^2 + z_1^2 z_3^2 + z_2^2 z_3^2.$$

$$\Lambda_r = \bigoplus_{\lambda, \lambda_1 \leq r} \mathbb{C} p_\lambda = \bigoplus_{\lambda, \lambda_1 \leq r} \mathbb{C} e_\lambda = \bigoplus_{\lambda, \ell(\lambda) \leq r} \mathbb{C} m_\lambda$$

Jack symmetric polynomials

Jack symmetric polynomials $P_\lambda^t(z)$ are uniquely characterised by

- 1 Upper triangular: $P_\lambda^t(z) = m_\lambda(z) + \sum_{\nu < \lambda} c_{\nu,\lambda}(t)m_\nu(z)$
- 2 Orthogonal: $\langle P_\lambda^t, P_\nu^t \rangle_r^t = 0$, if $\lambda \neq \nu$.

Well studied:

- $\langle P_\lambda^t(z), P_\lambda^t(z) \rangle_r^t = a_\lambda(t)$
- $\prod_{i,j}(1 - y_i z_j)^{-t} = \prod_{m \geq 1} \exp\left(t \frac{p_m(y)p_m(z)}{m}\right) = \sum_\nu b_\nu(t) P_\nu^t(y) P_\nu^t(z)$
- $\prod_i (1 + z_i)^{Xt} = \sum_\lambda d_\lambda(X, t) P_\lambda^t(z)$

where $a_\lambda(t)$, $b_\lambda(t)$, $c_{\nu,\lambda}(t)$, $d_\lambda(X, t)$ are rational functions in t . Explicit formulae known.

Examples:

- $P_{[1,\dots,1]}^t(z) = e_{[1,\dots,1]}(z)$
- $P_{[s,\dots,s]}^t(z_1, \dots, z_r) = \prod_{i=1}^r z_i^s, \ell([s, \dots, s]) = r$

Determining conformal weights from the zero mode

We want to evaluate the zero mode of the singular vector

$$S_+^{[p-1]}|(1-p)\alpha_+\rangle = \int_{\Gamma} S_+(z_1) \cdots S_+(z_{p-1})|(1-p)\alpha_+\rangle dz_1 \cdots dz_{p-1}.$$

The corresponding field is

$$\begin{aligned} S_+^{[p-1]}V_{(1-p)\alpha_+}(w) \\ = \int_{\Gamma} S_+(z_1 + w) \cdots S_+(z_{p-1} + w)V_{(1-p)\alpha_+}(w) dz_1 \cdots dz_{p-1}. \end{aligned}$$

The easiest way to evaluate such a zero mode is to evaluate matrix elements

$$\langle \tau | S_+^{[p-1]}V_{(1-p)\alpha_+}(w) | \tau \rangle,$$

where $\tau \in \mathbb{C}$ and $\langle \tau | a_{-n} = 0, n \geq 1$.

Determining conformal weights from the zero mode

$$\langle \tau | S_+^{[p-1]} V_{(1-p)\alpha_+}(w) | \tau \rangle = \left\langle P_{[(q-1)^{p-1}]}^{\frac{\alpha_+^2}{2}}(z), \prod_{i=1}^{p-1} \left(1 + \frac{z_i}{w}\right)^{\alpha_+ \tau} \right\rangle_{p-1}^{\alpha_+^2/2}$$

There are known formulae for expanding

$$\prod_{i=1}^{p-1} \left(1 + \frac{z_i}{w}\right)^{\alpha_+ \tau} = \sum_{\nu} g_{\nu}(\alpha_+ \tau) P_{\nu}^{\frac{\alpha_+^2}{2}}(z) w^{-|\nu|}.$$

Determining conformal weights from the zero mode

$$\langle \tau | S_+^{[p-1]} V_{(1-p)\alpha_+}(w) | \tau \rangle \propto g_{[(q-1)p-1]}(\alpha_+ \tau) \propto \prod_{r,s} (h_\tau - h_{r,s}),$$

where $1 \leq r \leq p-1$, $1 \leq s \leq q-1$, $(r,s) \simeq (p-r, q-s)$ and

$$h_\tau = \frac{\tau}{2}(\tau - \beta), \quad h_{r,s} = \frac{(qs - pr)^2 - (p - q)^2}{4pq}.$$

Theorem

Let $f(h_\tau) = \prod_{r,s} (h_\tau - h_{r,s})$.

Zhu's algebra: $\text{Zhu}(\mathbf{M}(p, q)) = \frac{\mathbb{C}[x]}{\langle f(x) \rangle}$

The minimal model highest (conformal) weights are the zeros of $f(x)$.

General scheme

- Pick a universal vertex operator algebra V
 - universal: finite number of generators and OPE relations, freely generated by derivatives and normal ordering.
 - come in families parametrised by central elements (central charge, level, etc.)
 - generically simple. Minimal models are the simple quotients by non-trivial ideals.
- Find a free field realisation: $V \hookrightarrow F$
 - Find screening fields: Fields whose OPEs with V are total derivatives.
 - Construct the ideal of V as the image of screening operators
- Compute Zhu's algebra
 - Zhu's algebra of V is generated by the image of the generators of V .
 - Compute ideal in $\text{Zhu}(V)$ using screening operators and symmetric polynomials $\Rightarrow$ Zhu algebra of minimal models.

General scheme

- Pick a universal vertex operator algebra V
 - universal: finite number of generators and OPE relations, freely generated by derivatives and normal ordering.
 - come in families parametrised by central elements (central charge, level, etc.)
 - generically simple. Minimal models are the simple quotients by non-trivial ideals.
- Find a free field realisation: $V \hookrightarrow F$
 - Find screening fields: Fields whose OPEs with V are total derivatives.
 - Construct the ideal of V as the image of screening operators
- Compute Zhu's algebra
 - Zhu's algebra of V is generated by the image of the generators of V .
 - Compute ideal in $\text{Zhu}(V)$ using screening operators and symmetric polynomials $\Rightarrow$ Zhu algebra of minimal models.

General scheme

- Pick a universal vertex operator algebra V
 - universal: finite number of generators and OPE relations, freely generated by derivatives and normal ordering.
 - come in families parametrised by central elements (central charge, level, etc.)
 - generically simple. Minimal models are the simple quotients by non-trivial ideals.
- Find a free field realisation: $V \hookrightarrow F$
 - Find screening fields: Fields whose OPEs with V are total derivatives.
 - Construct the ideal of V as the image of screening operators
- Compute Zhu's algebra
 - Zhu's algebra of V is generated by the image of the generators of V .
 - Compute ideal in $\text{Zhu}(V)$ using screening operators and symmetric polynomials $\Rightarrow$ Zhu algebra of minimal models.

General scheme

Scheme works for:

- Virasoro
- affine $\mathfrak{sl}(2)$
- $N = 1$ Virasoro

Seems to work for:

- affine $\mathfrak{osp}(1|2)$

Should also work for higher ranks:

- affine $\mathfrak{sl}(3)$ or any affinisation of a simple Lie algebra
- affine $\mathfrak{sl}(2|1)$ or any affinisation of a simple Lie superalgebra
- $N = 2$ Virasoro, etc.

The End

Thank you for coming!